

KOINGNAAS, South Africa

WMO: 685130

Lat: 30.1956S

Lon: 17.2903E

Elev: 312

StdP: 14.53

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	41.6	43.7	19.9	15.1	70.3	24.6	18.9	69.7	26.7	60.2	24.7	63.3	9.0	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	88.7	62.3	84.5	60.9	80.7	60.3	68.2	77.7	66.8	75.4	65.8	73.6	13.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	95.2	70.6	64.0	90.4	69.4	63.0	87.0	68.8	32.7	78.0	31.5	75.5	30.8	73.8	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	25.2	22.8	DB	37.2	97.2	1.7	1.8	36.0	98.5	35.0	99.5	34.0	100.5	32.8	101.9	(4)
(5)			WB	35.5	69.6	2.1	2.2	34.0	71.2	32.8	72.4	31.7	73.7	30.2	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	68.0	67.9	66.4	64.7	60.9	58.3	57.8	56.3	59.0	61.8	64.1	66.5	(6)	
	DBStd	6.39	2.56	3.01	4.37	6.47	6.24	6.38	7.23	5.11	4.73	4.52	3.68	2.75	(7)	
	HDD50	6	0	0	0	0	0	1	4	1	0	0	0	0	(8)	
	HDD65	1430	4	5	26	80	166	228	254	282	195	121	56	13	(9)	
	CDD50	4608	558	501	509	440	337	249	247	196	269	367	423	512	(10)	
	CDD65	556	96	85	70	70	39	26	32	12	14	23	29	60	(11)	
	CDH74	3022	212	260	369	571	319	192	205	108	169	253	184	180	(12)	
	CDH80	945	25	71	135	213	101	57	45	33	64	99	58	44	(13)	
(14) Wind	WSAvg	11.3	12.9	12.1	11.4	10.2	9.3	9.8	9.9	10.3	11.5	12.2	12.7	13.1	(14)	
(15) Precipitation	PrecAvg	4.8	0.2	0.1	0.2	0.3	0.5	0.7	0.7	0.8	0.4	0.3	0.4	0.1	(15)	
	PrecMax	6.6	0.6	0.4	0.8	0.9	1.6	2.0	2.4	3.2	1.7	1.3	2.6	0.4	(16)	
	PrecMin	2.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	(17)	
	PrecStd	1.1	0.2	0.1	0.2	0.3	0.4	0.6	0.6	0.7	0.4	0.3	0.6	0.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.2	89.0	90.9	92.2	88.4	86.4	84.9	85.5	88.2	91.1	86.9	87.4	(19)	
		MCWB	66.7	67.0	65.4	64.1	60.0	59.0	57.9	56.6	57.2	60.0	61.1	66.2	(20)	
	2%	DB	78.1	80.2	84.0	86.2	81.9	79.9	79.9	76.4	79.0	82.0	78.8	77.7	(21)	
		MCWB	65.7	65.7	63.3	61.7	57.8	55.9	56.3	54.6	56.6	59.7	60.6	63.8	(22)	
	5%	DB	75.9	76.3	77.4	80.6	76.3	73.9	74.9	69.2	71.8	74.4	73.8	75.0	(23)	
		MCWB	64.7	64.9	63.4	60.3	56.7	54.3	54.2	52.0	54.6	58.6	60.5	63.3	(24)	
	10%	DB	74.1	74.3	73.6	75.4	71.1	68.8	69.5	64.3	66.6	69.4	70.8	72.9	(25)	
		MCWB	64.1	64.1	63.0	59.0	55.8	53.0	52.7	52.6	55.4	57.8	59.8	62.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	70.4	69.3	68.0	62.8	60.9	59.9	59.3	62.6	65.6	65.4	68.5	(27)	
		MCDB	76.9	80.6	80.1	85.1	72.0	75.2	75.2	71.7	74.5	81.1	75.3	77.7	(28)	
	2%	WB	67.9	67.6	67.1	65.2	61.4	58.4	58.0	57.1	59.4	62.5	63.4	66.2	(29)	
		MCDB	74.6	75.8	76.9	79.4	70.5	67.2	69.9	66.0	69.9	72.1	72.0	74.2	(30)	
	5%	WB	66.5	66.4	65.5	63.0	60.1	57.3	56.6	55.8	57.8	60.8	62.2	65.1	(31)	
		MCDB	72.9	74.0	73.9	73.6	68.3	64.9	66.8	63.1	66.6	68.7	70.2	72.3	(32)	
	10%	WB	65.5	65.4	64.2	61.7	58.8	56.2	55.3	54.8	56.6	59.4	61.1	64.1	(33)	
		MCDB	71.7	72.3	71.6	70.4	66.1	62.9	64.1	61.0	64.0	66.6	68.3	70.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	14.4	15.2	16.7	17.4	17.6	18.3	16.4	16.0	15.9	15.1	14.2	(35)	
		MCDBR	16.2	17.9	21.5	24.9	26.0	26.1	25.5	26.8	26.3	27.4	21.5	17.2	(36)	
		MCWBR	6.8	7.2	8.5	9.6	11.0	11.1	10.6	12.0	11.1	11.2	8.9	7.3	(37)	
	5% WB	MCDBR	13.9	15.0	16.0	18.3	18.1	18.2	20.0	18.0	18.3	17.9	16.8	14.2	(38)	
		MCWBR	7.0	7.2	8.0	9.1	10.1	10.7	11.2	10.3	9.6	9.2	8.3	7.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.345	0.334	0.317	0.301	0.292	0.300	0.317	0.333	0.343	0.328	0.332	(40)	
	taud		2.523	2.519	2.542	2.589	2.602	2.584	2.556	2.504	2.425	2.427	2.523	2.526	(41)	
	Ebn at Noon		318	310	303	293	283	278	280	288	298	307	319	321	(42)	
	Edh at Noon		36	35	32	27	24	23	25	29	35	37	35	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2640	2410	2036	1556	1188	1028	1126	1380	1792	2233	2565	2730	(44)	
	RadStd		94	80	58	60	55	51	48	56	74	66	81	102	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page